高等影像處理

作業#3

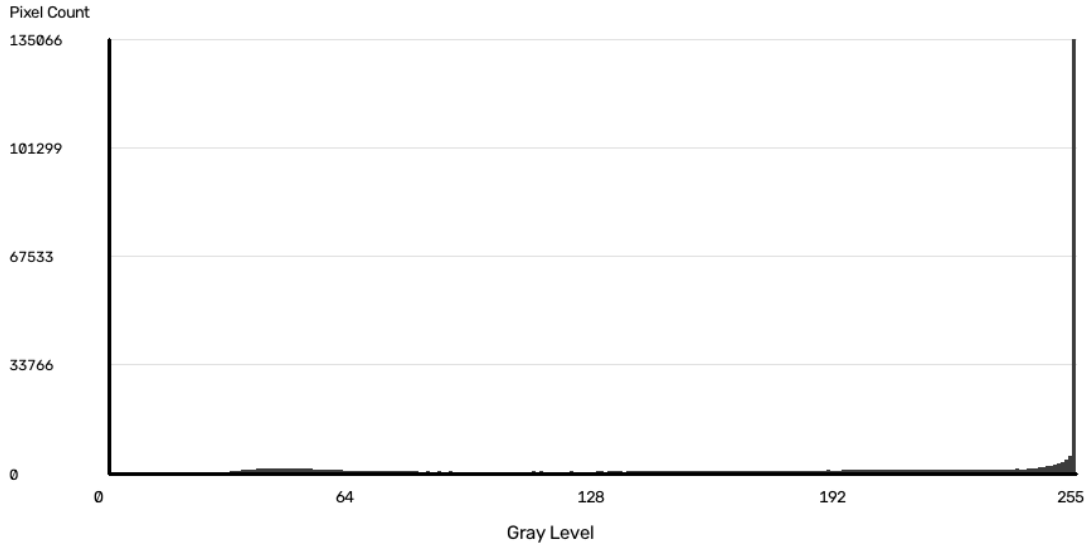
姓　　名：王 定 亮

學　　號：115318056

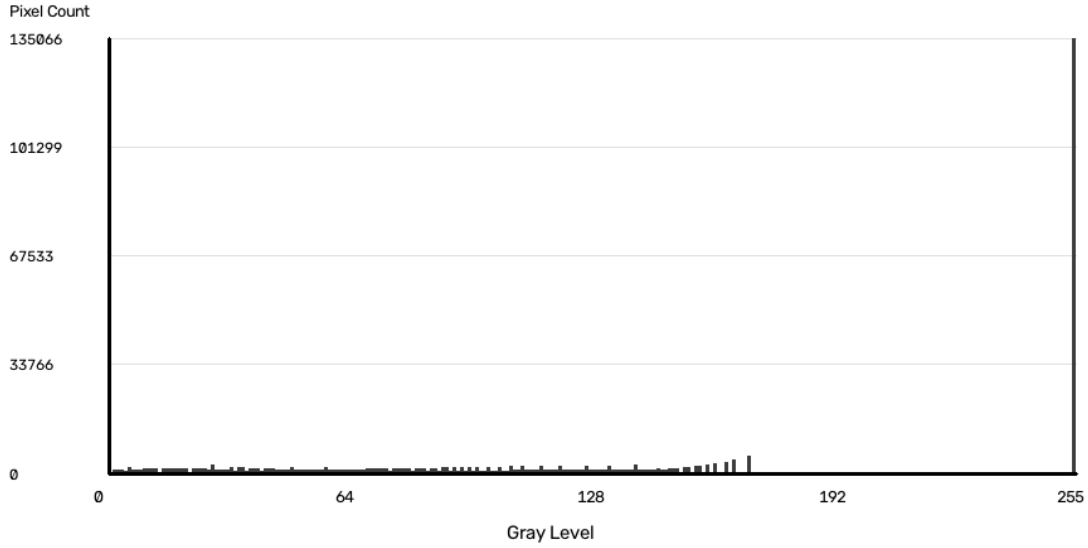
指導老師：黃 正 民

1.1

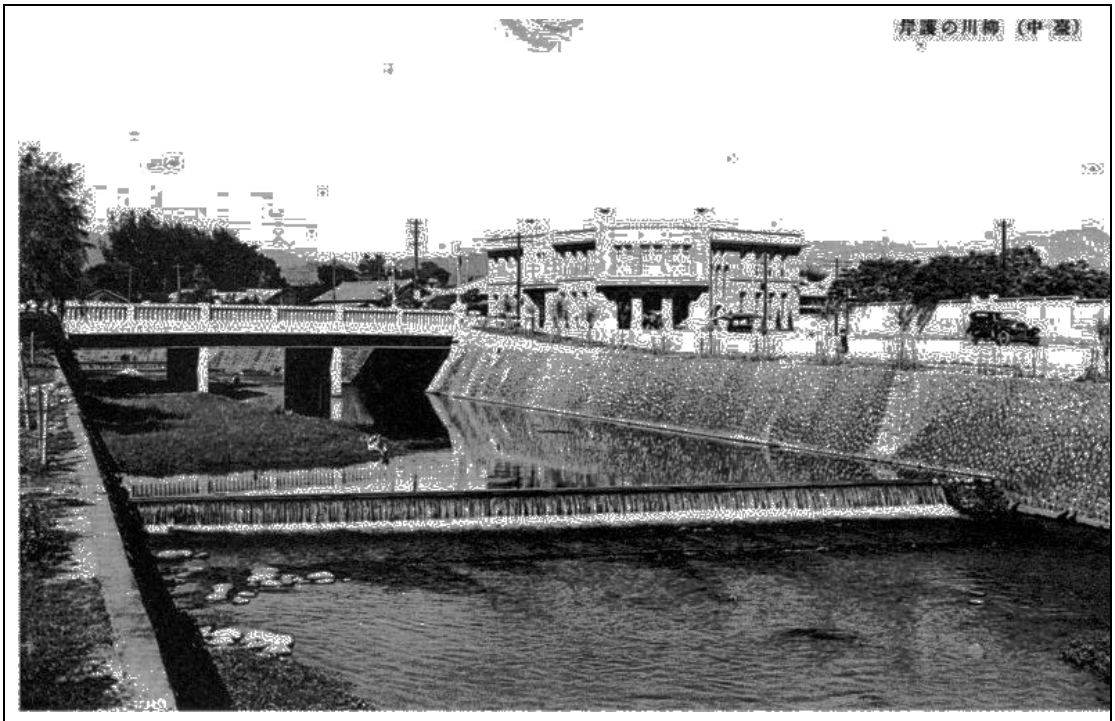
Figure



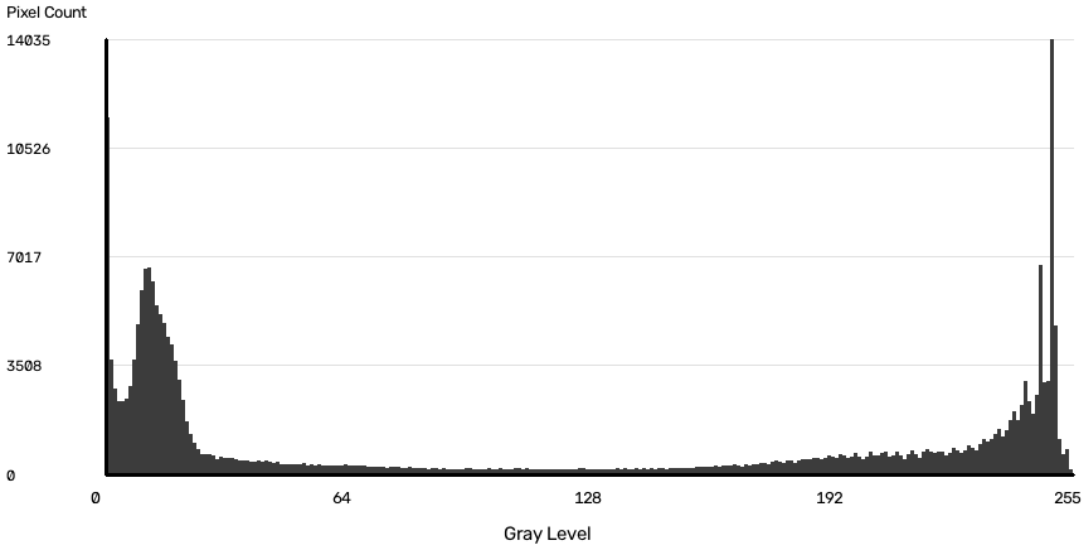
Liuchuan800x500_hist_before.png



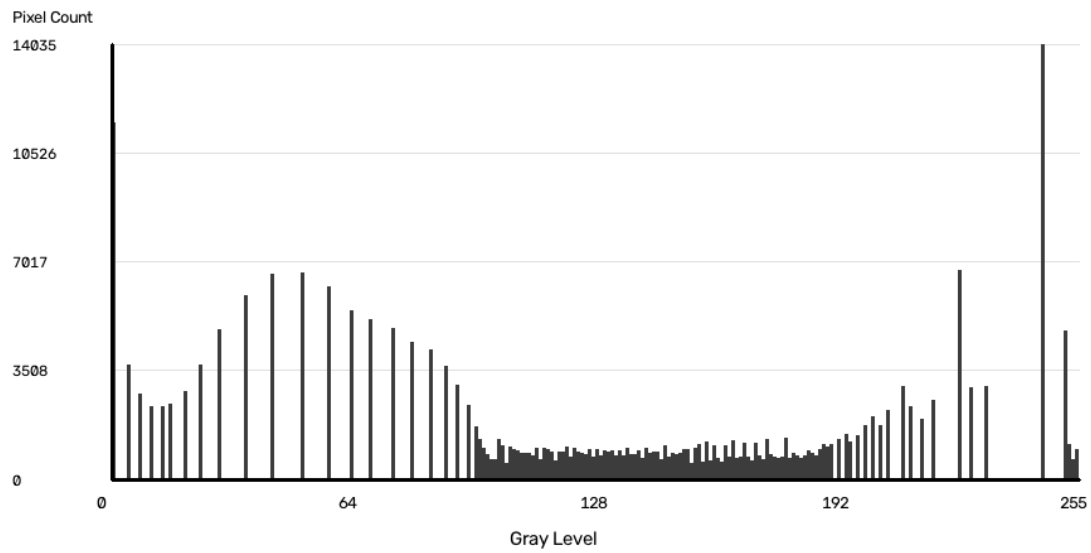
Liuchuan800x500_hist_after.png



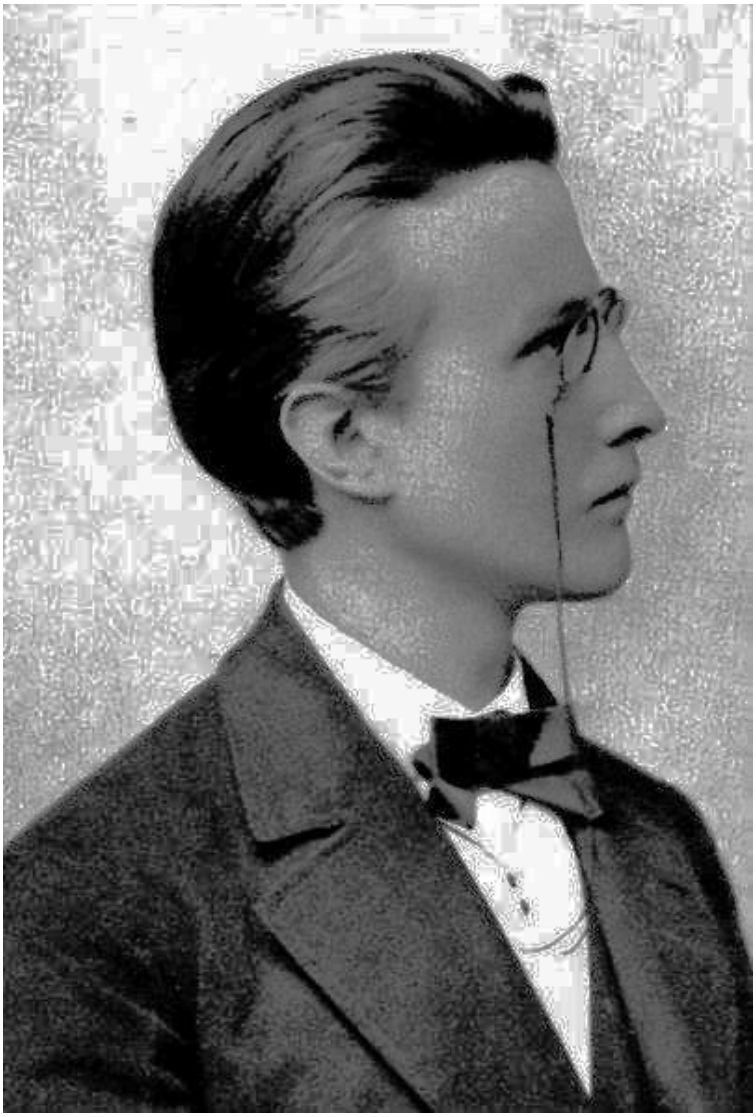
Liuchuan800x500_equalized.png



Planck403x596_hist_before.png



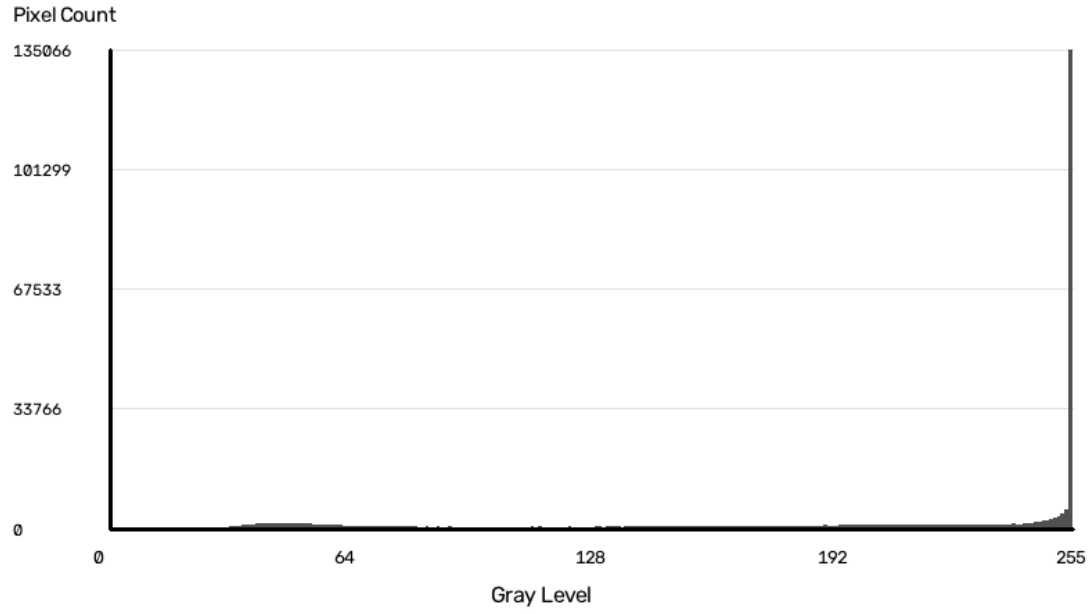
Planck403x596_hist_after.png



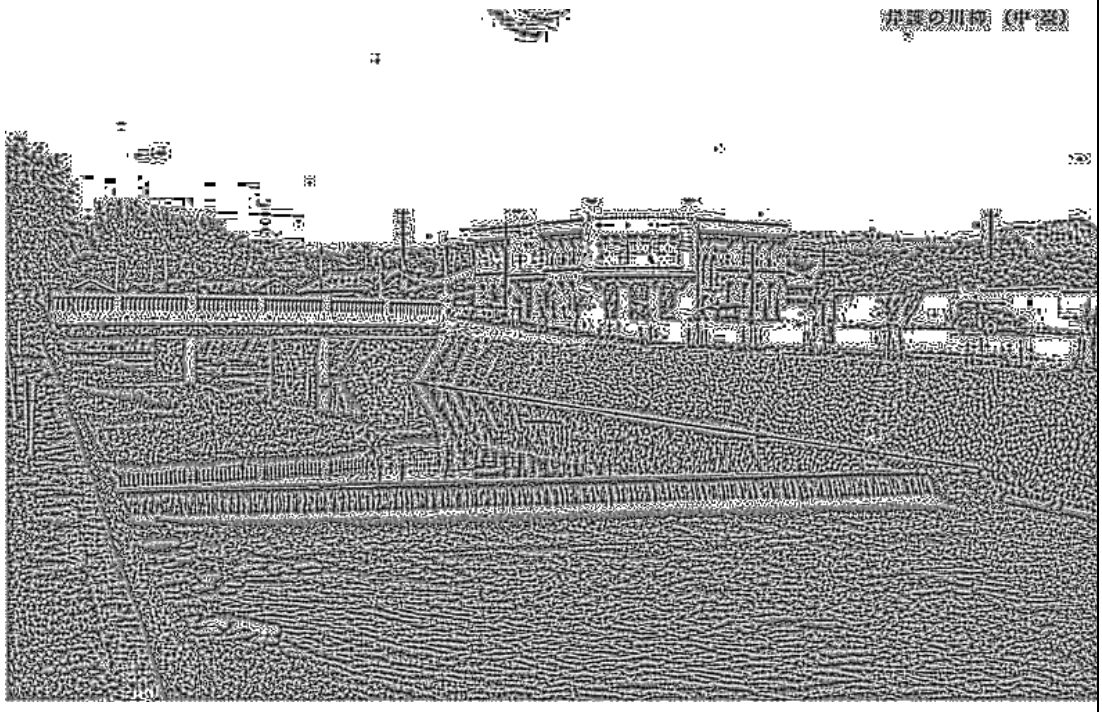
Planck403x596_equalized.png

1.2

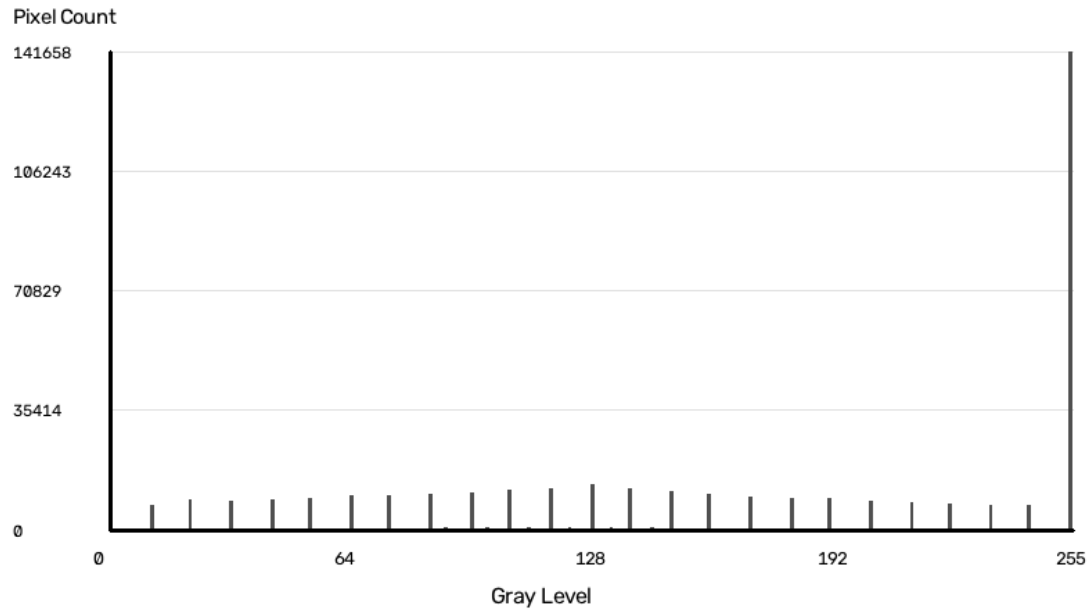
Figure



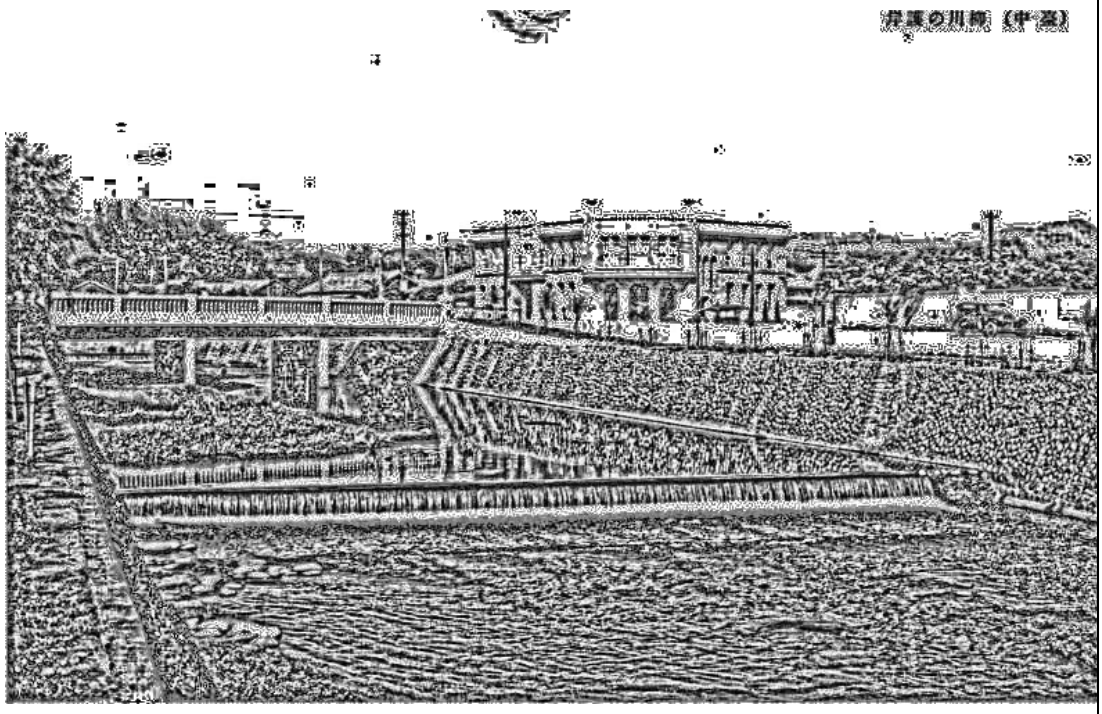
Liuchuan800x500_hist_before.png



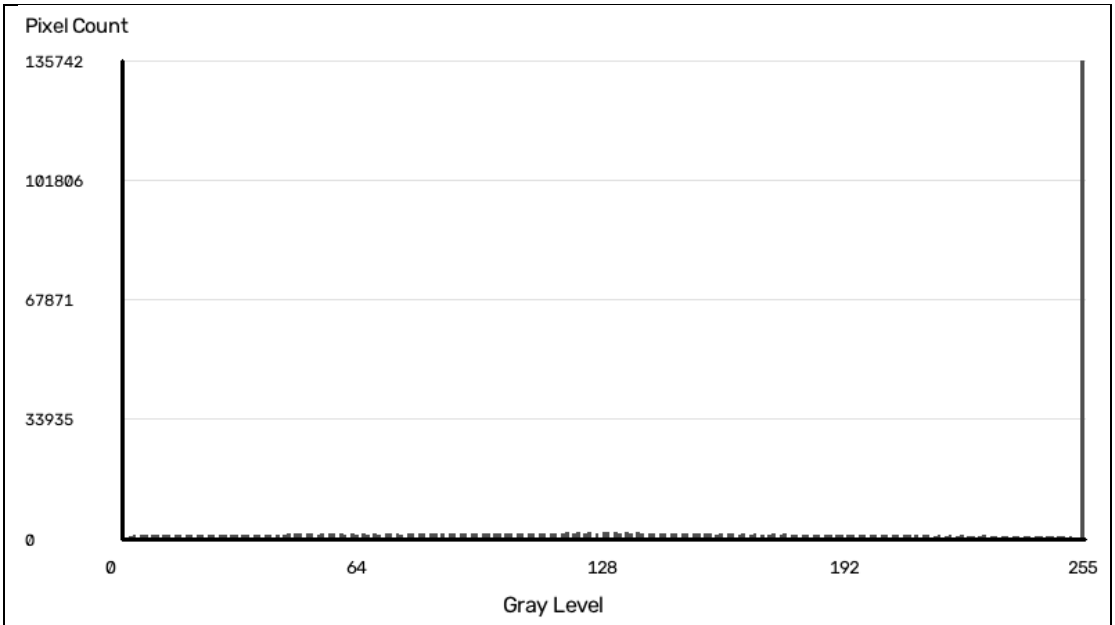
Liuchuan800x500_local_5x5.png



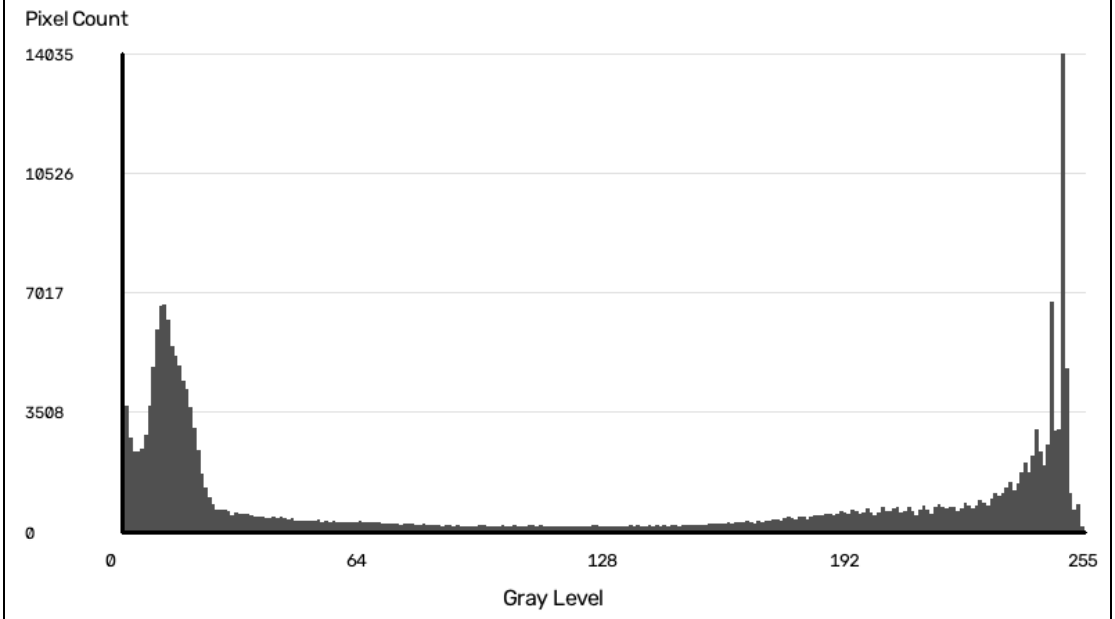
Liuchuan800x500_local_5x5_hist.png



Liuchuan800x500_local_13x13.png



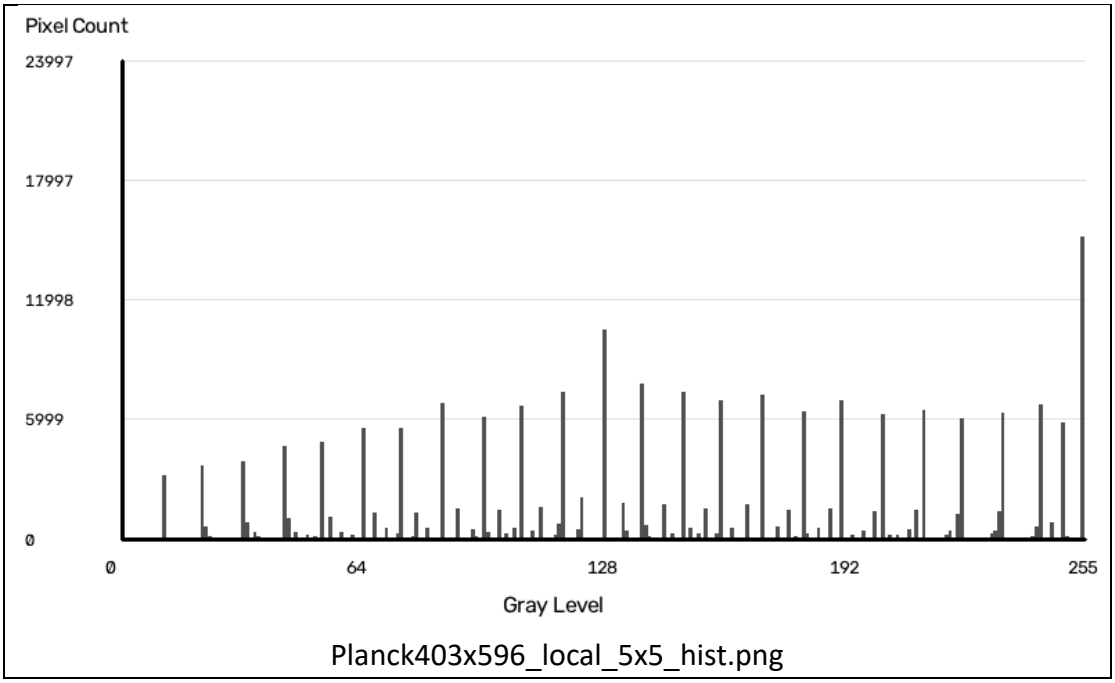
Liuchuan800x500_local_13x13_hist.png



Planck403x596_hist_before.png



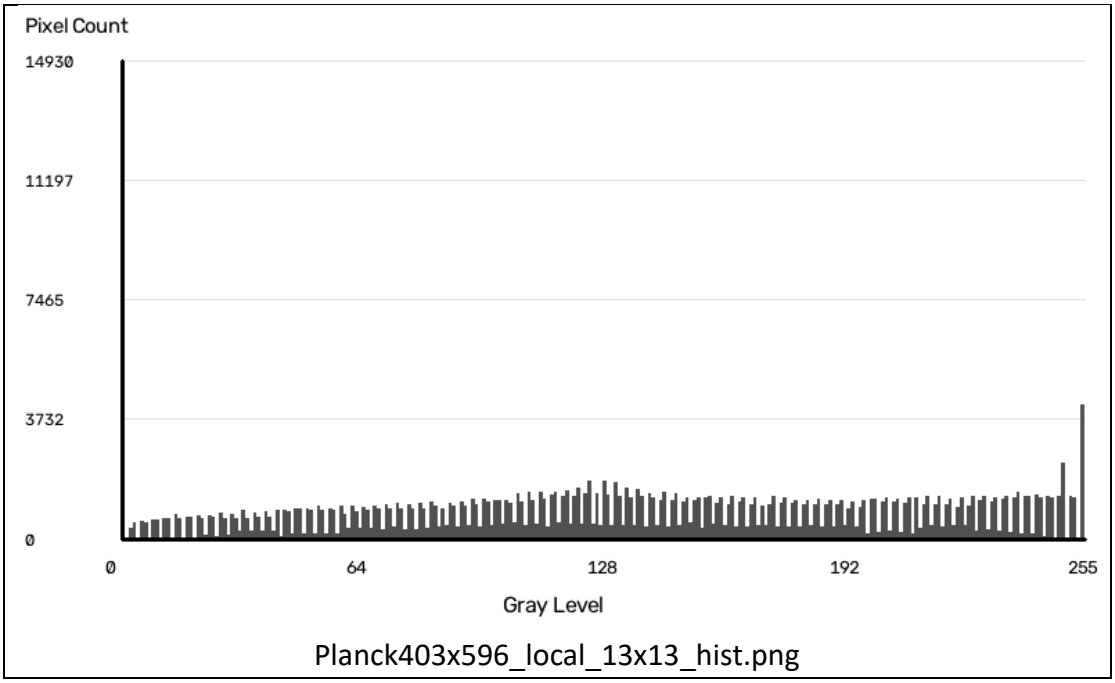
Planck403x596_local_5x5.png



Planck403x596_local_5x5_hist.png



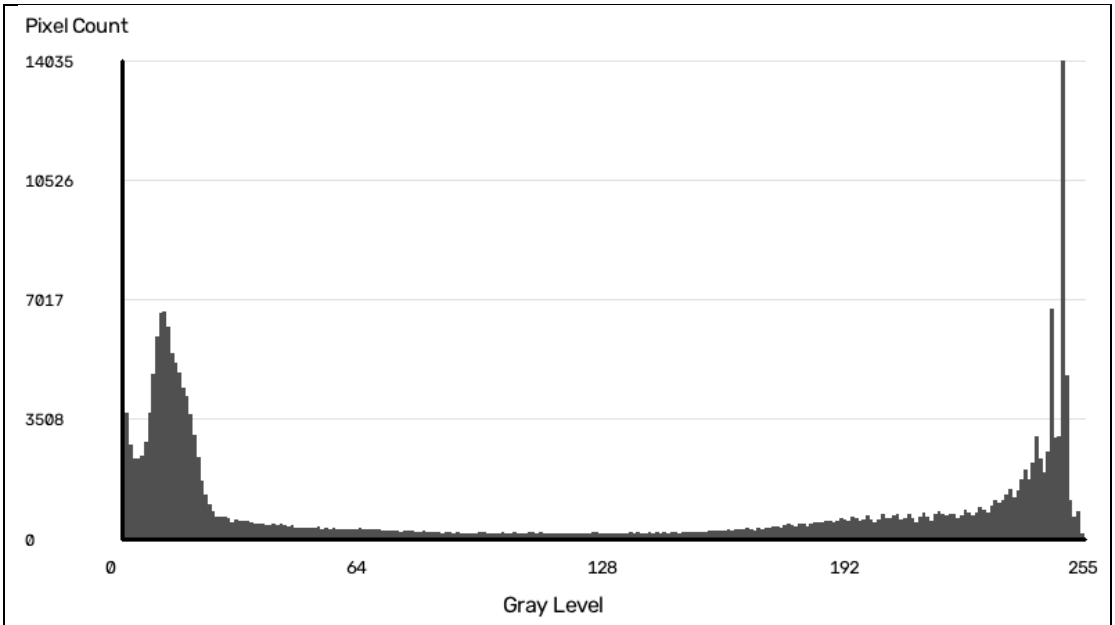
Planck403x596_local_13x13.png



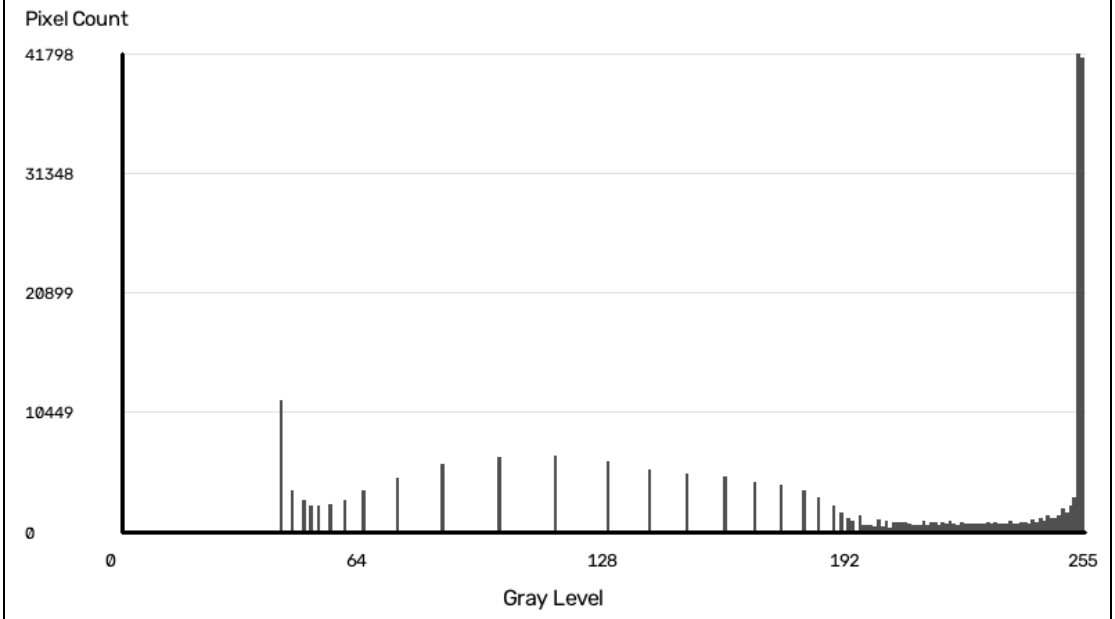
Planck403x596_local_13x13_hist.png

1.3
Figure

Planck_matched.png



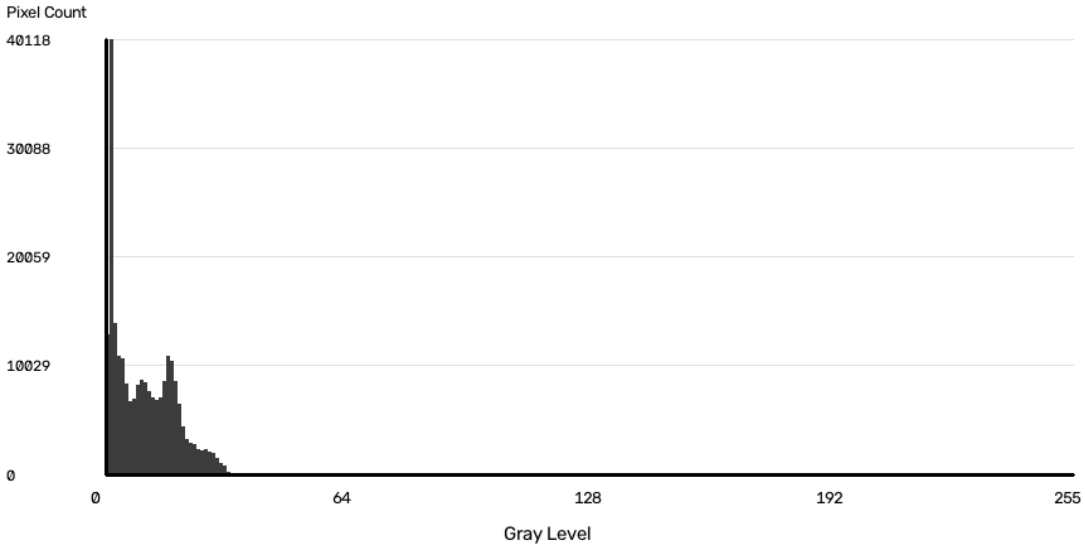
Planck_hist_before.png



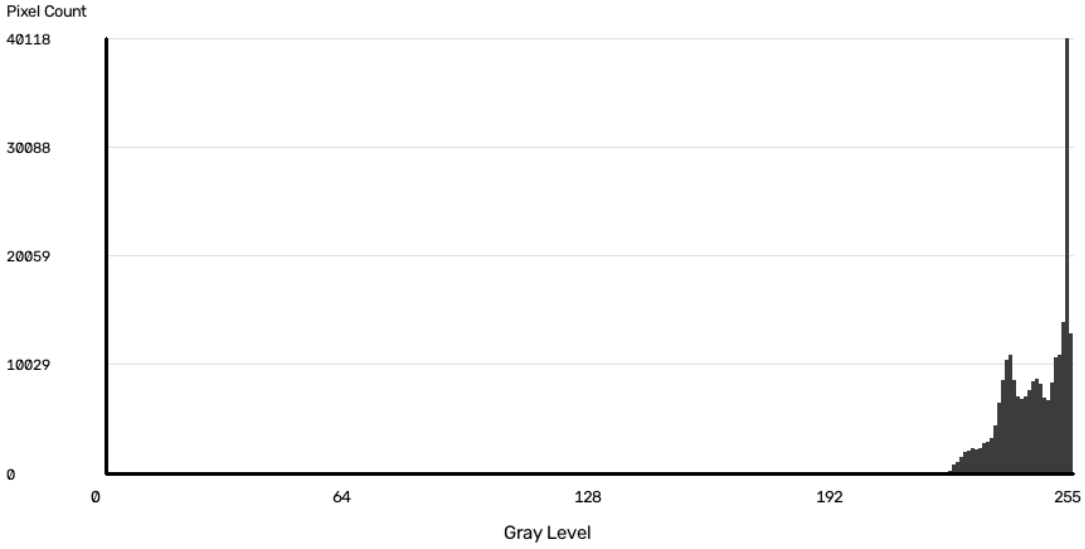
Planck_hist_matched.png

1.4
Disgussion
可以發現上面三種效果其實最終的結果都不好，全域的作法會導致背景雜訊被放大，local 的做法則會導致平面區域出現大量噪聲與光暈，第三種做法，兩張圖之間完全沒關係，自然效果也不好

2.1
Figure



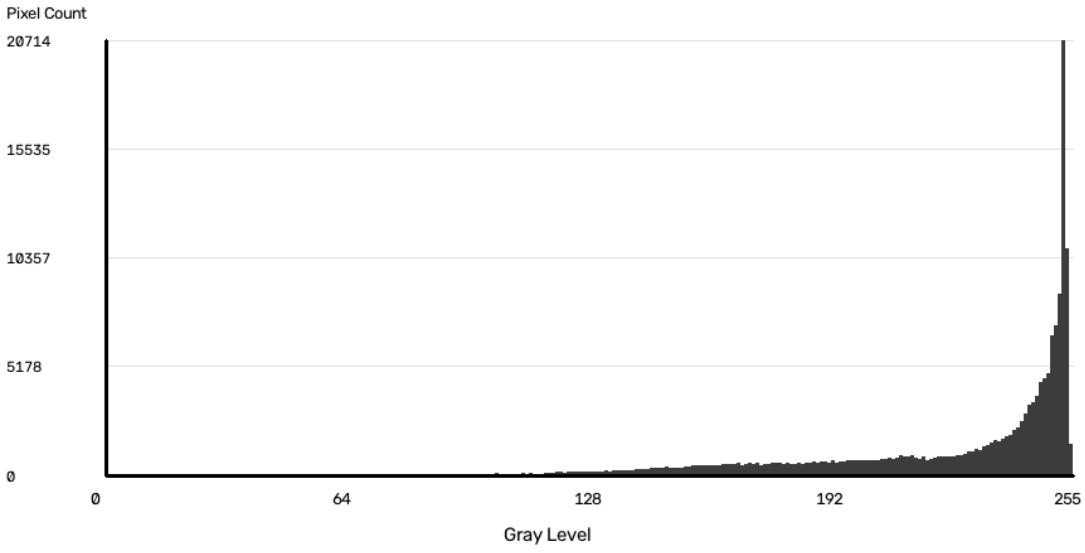
dark600x400_hist_before.png



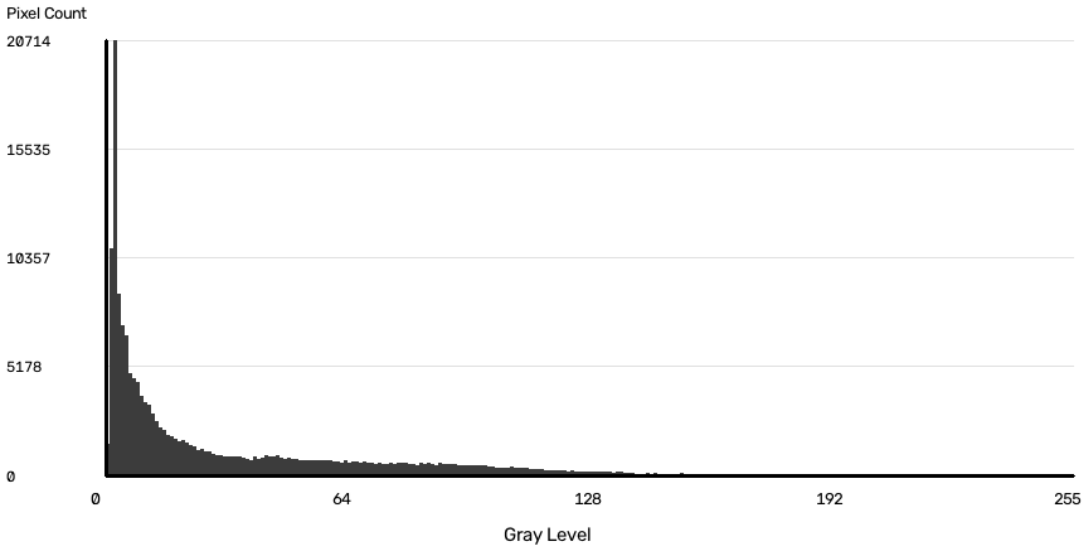
dark600x400_hist_after.png



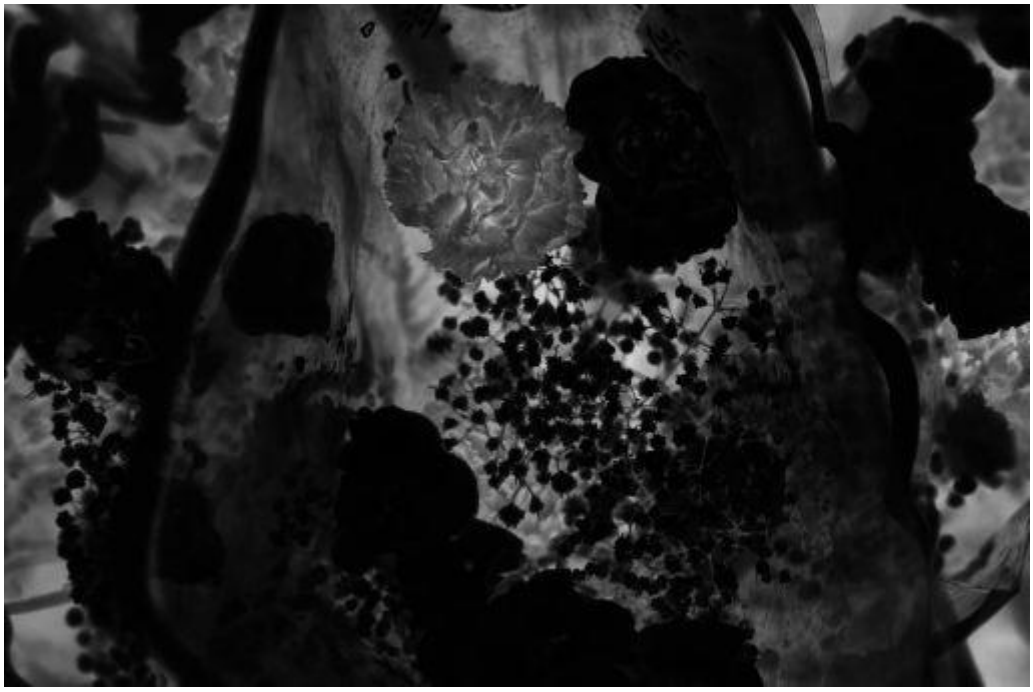
dark600x400_negative.png



light512x341_hist_before.png



light512x341_hist_after.png



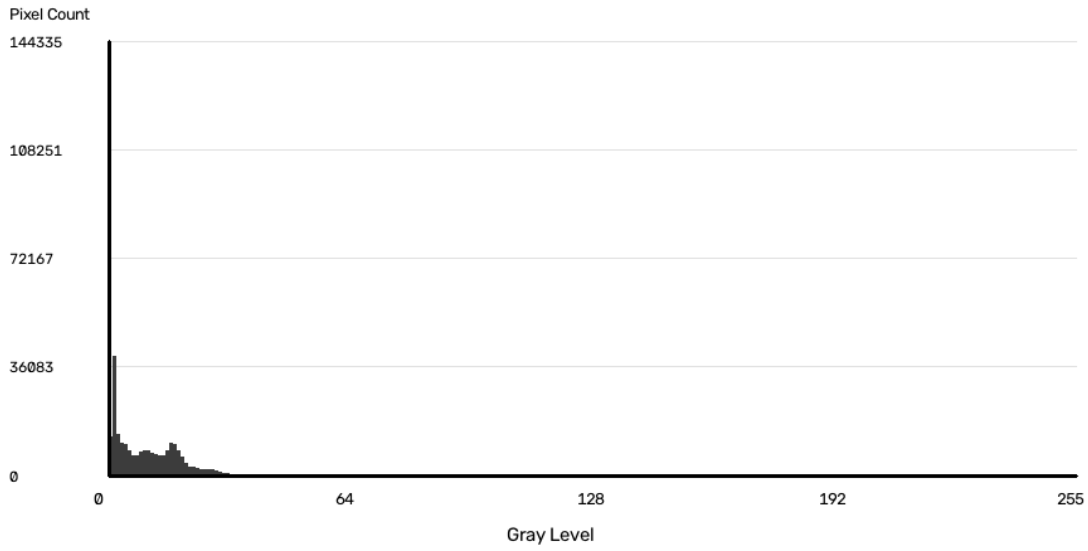
light512x341_negative.png

Disgussion

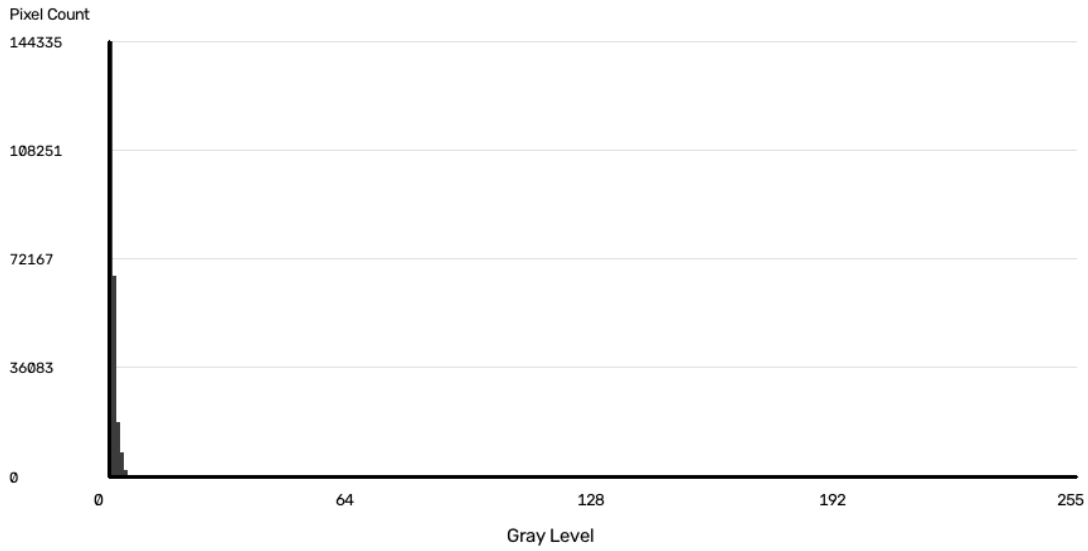
負片效果就是把亮的地方換成暗的，暗的換成亮的，原本很亮的就變很暗，很暗的就變很亮，結果還看效果並不好

2.2

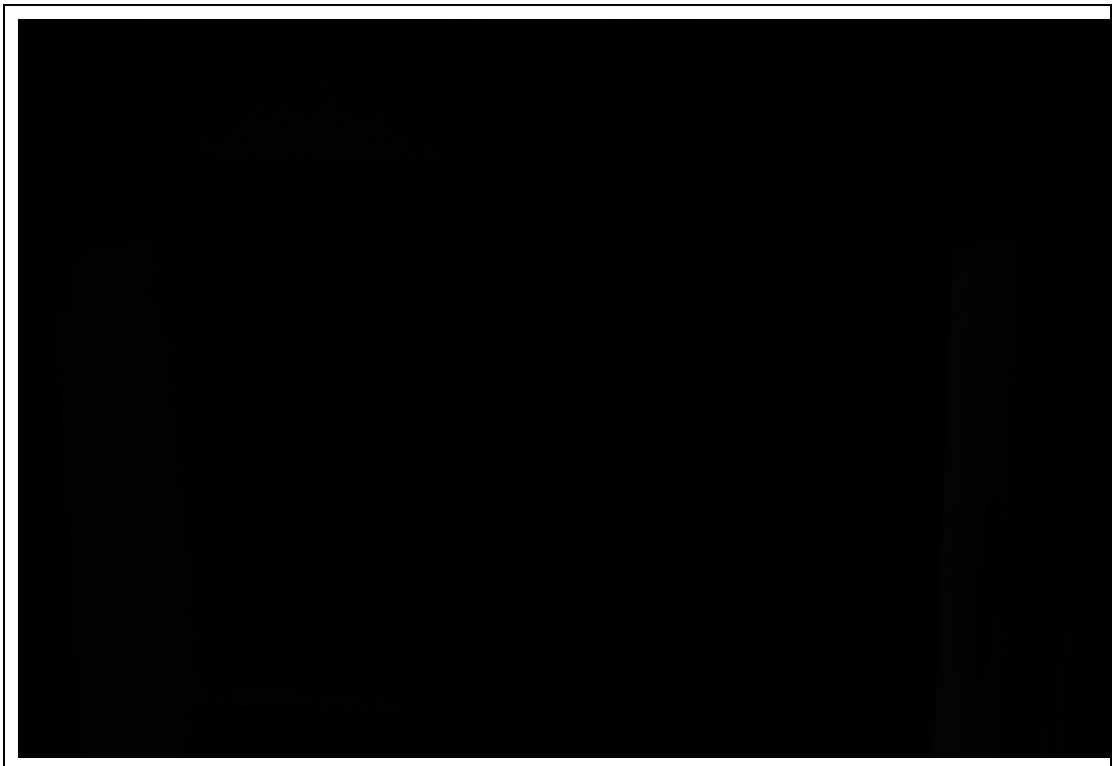
Figure



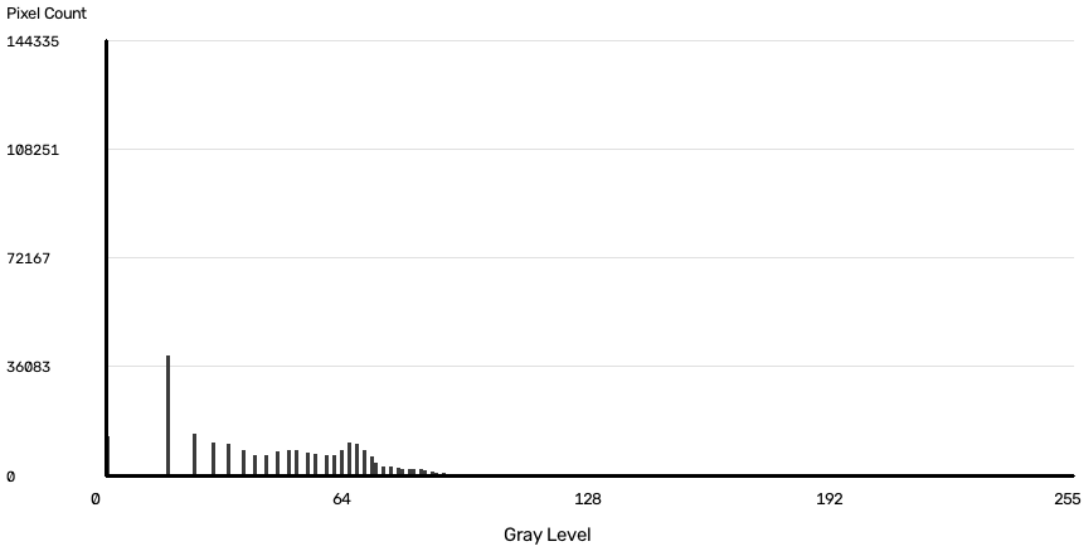
dark600x400_hist_before.png



dark600x400_gamma_2p0_hist.png



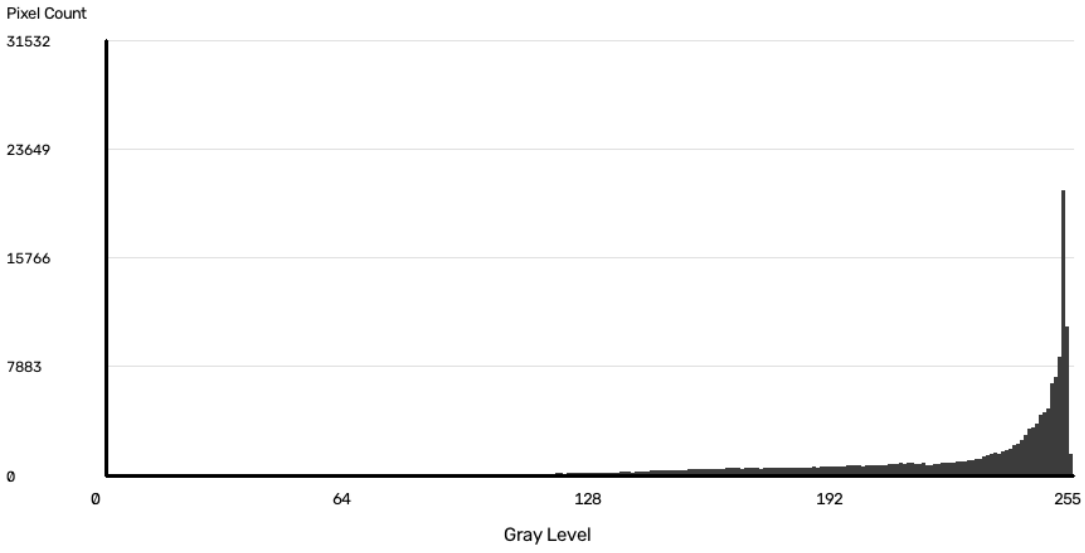
dark600x400_gamma_2p0.png



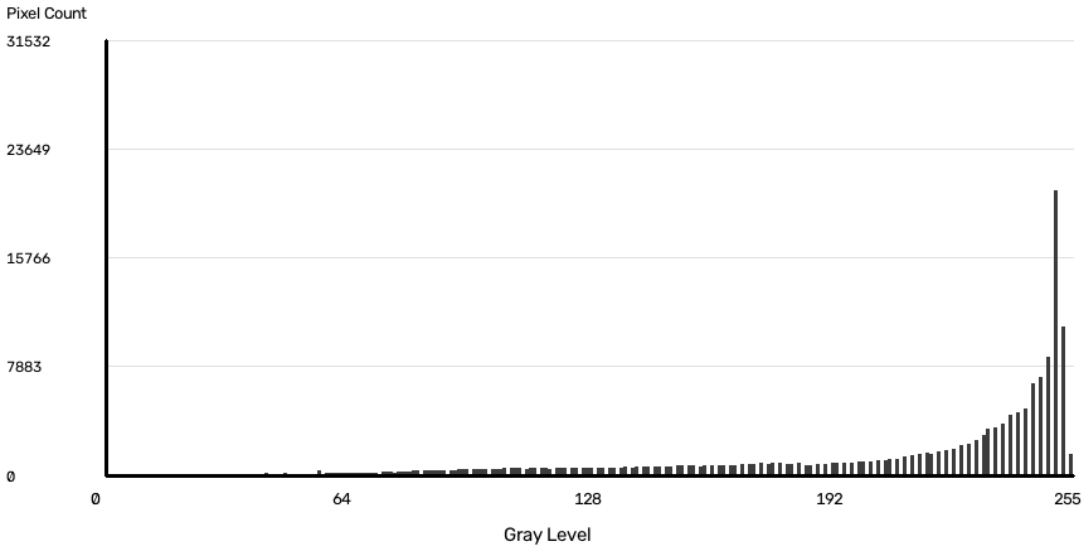
dark600x400_gamma_0p5_hist.png



dark600x400_gamma_0p5.png



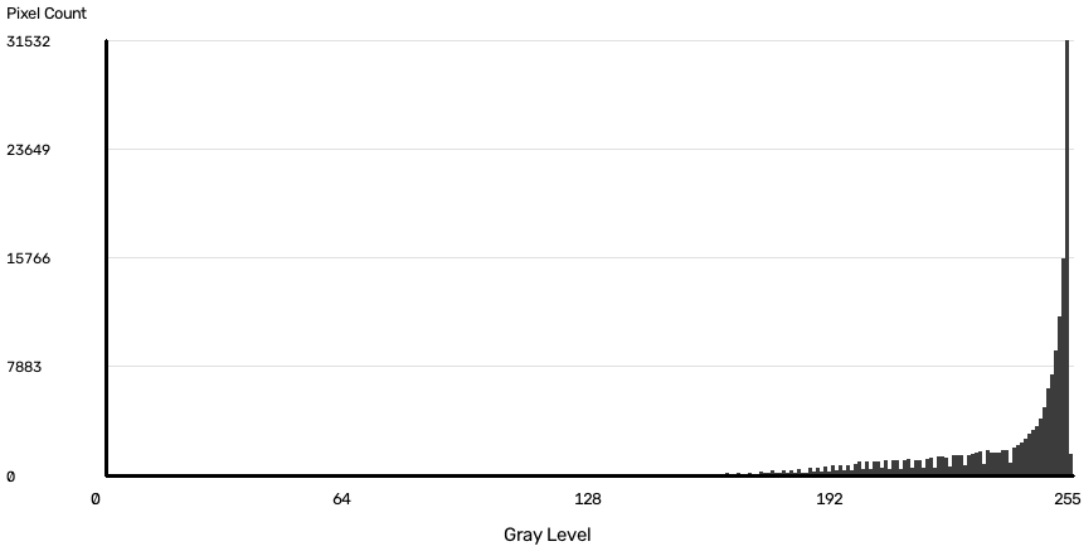
light512x341_hist_before.png



light512x341_gamma_2p0_hist.png



light512x341_gamma_2p0.png



light512x341_gamma_0p5_hist.png

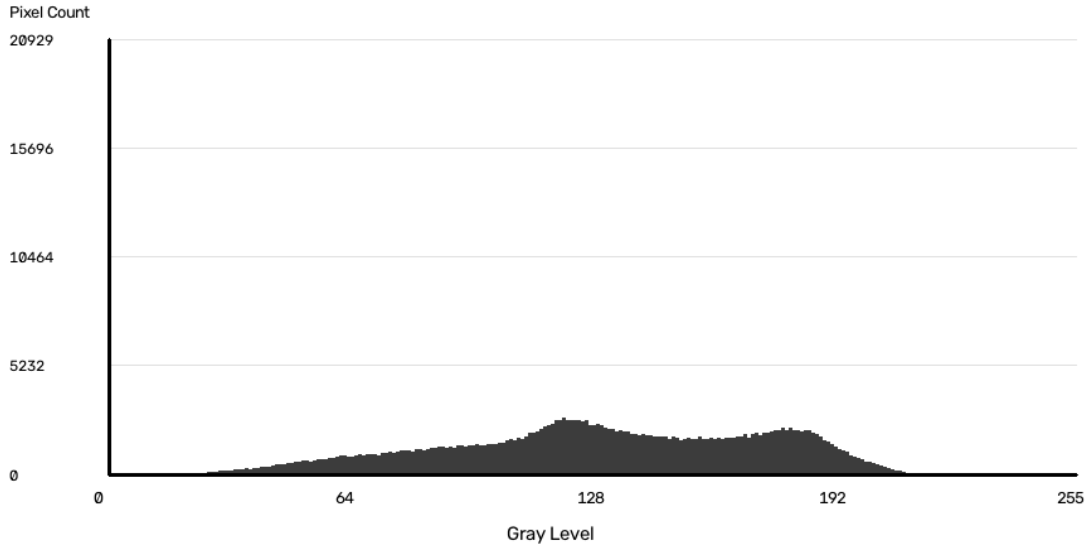


light512x341_gamma_0p5.png

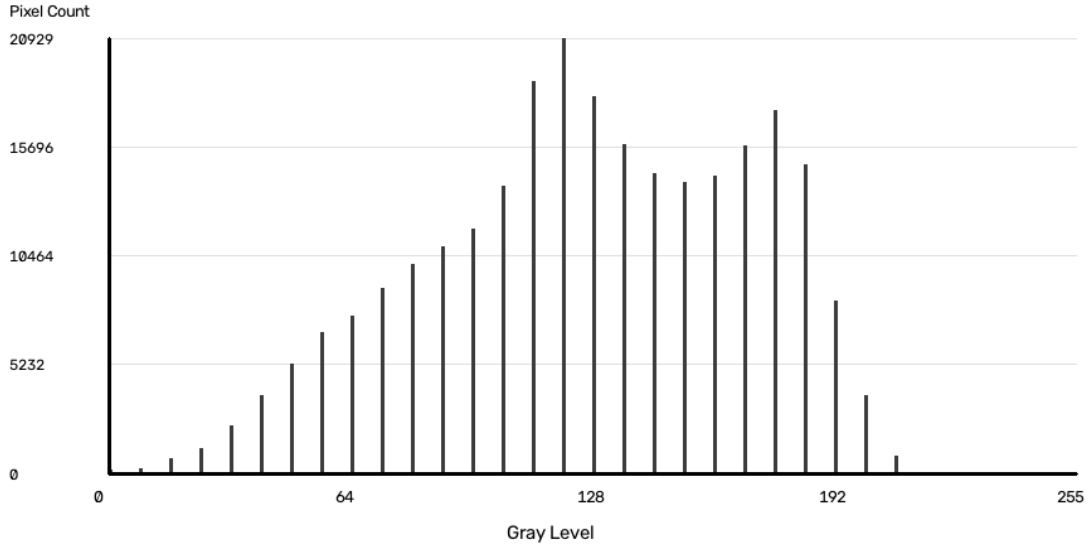
Disgussion

這題 gamma 我選的是 2.0 和 0.5，該值越小越往亮的地方強化，反之則往暗的地方強化，觀察可以發現將較暗的圖片經過 Power-law gamma0.5 強化過後，櫃子裡的東西清楚不少

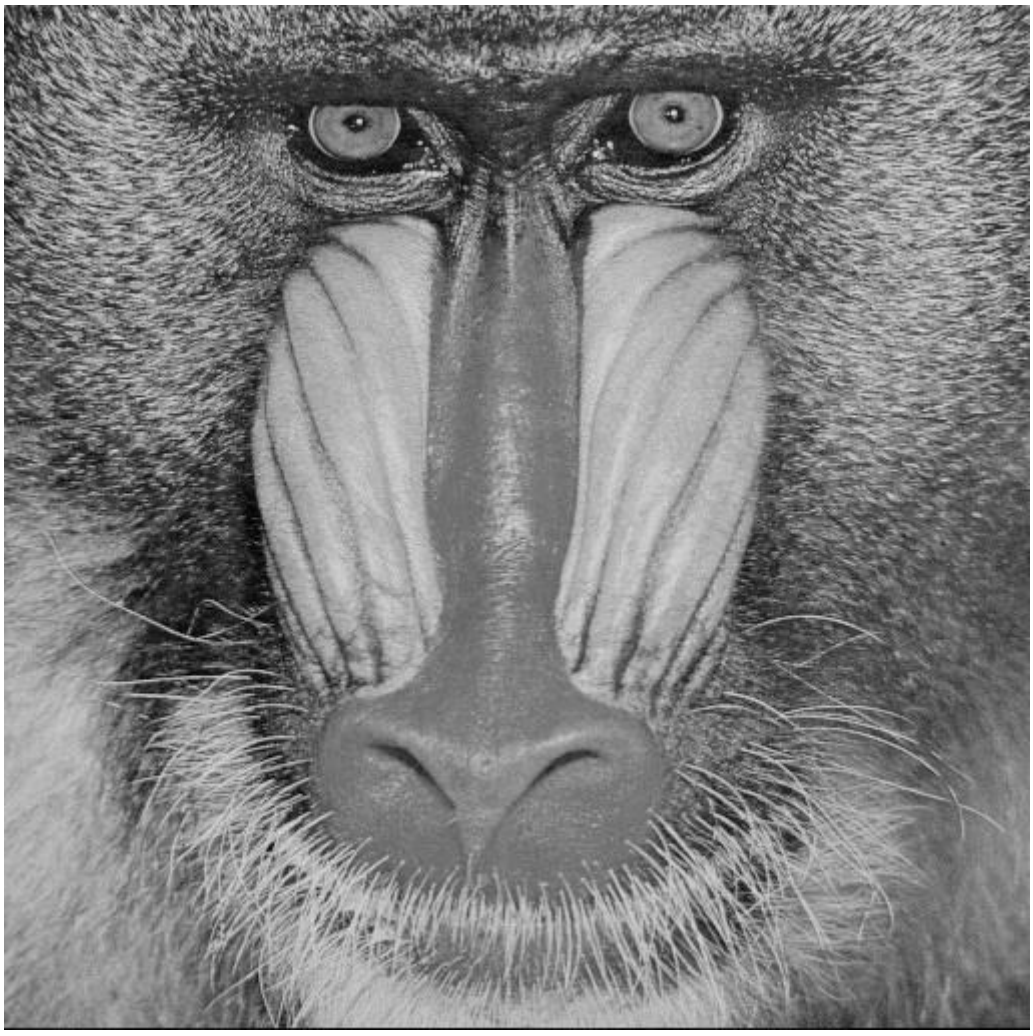
3.1
Figure



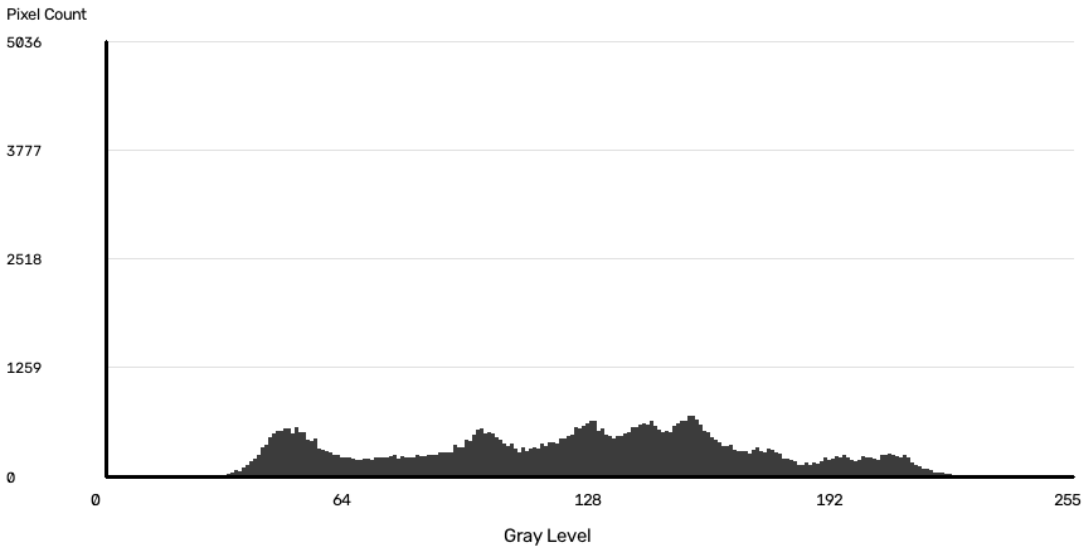
baboon512_hist_before.png



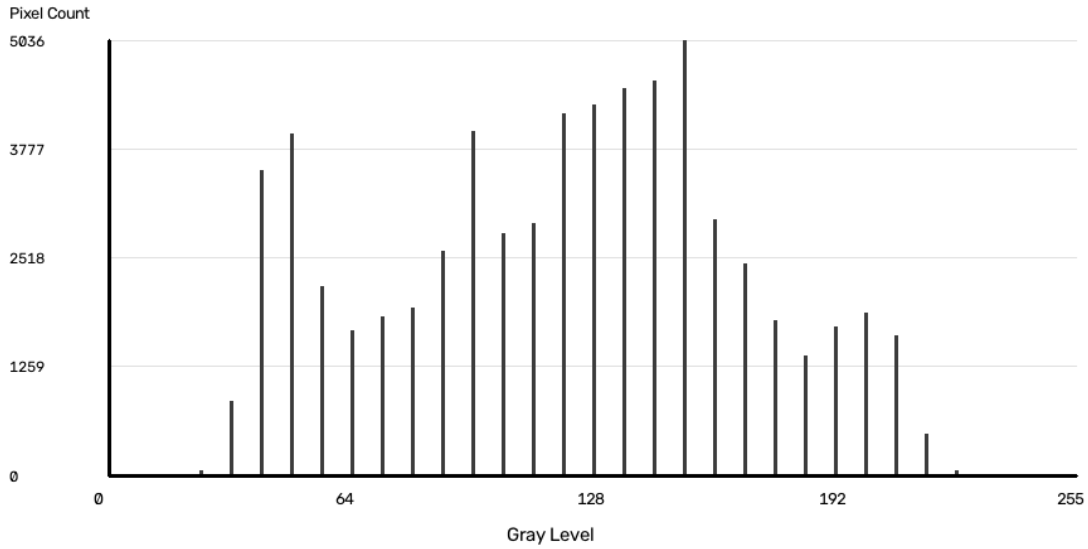
baboon512_hist_after.png



baboon512_msb_reconstructed.png



lena256_hist_before.png



lena256_hist_after.png

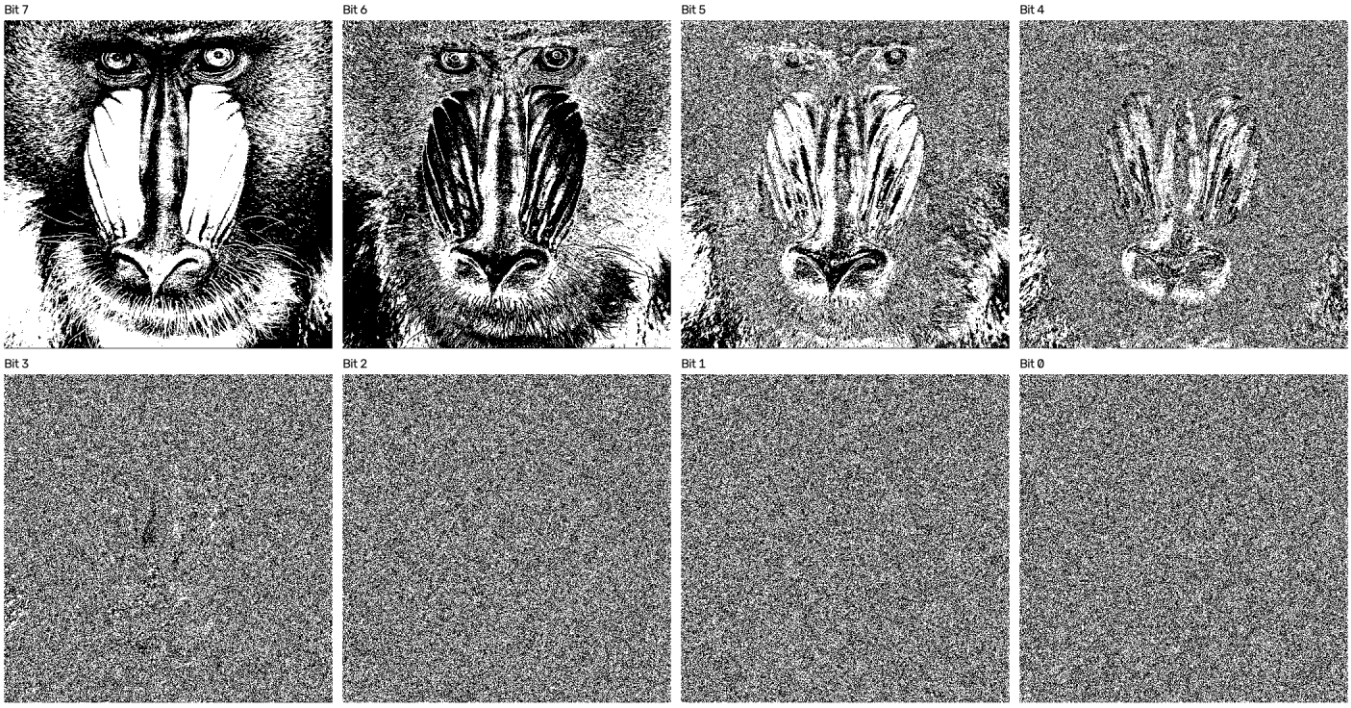


lena256_msb_reconstructed.png

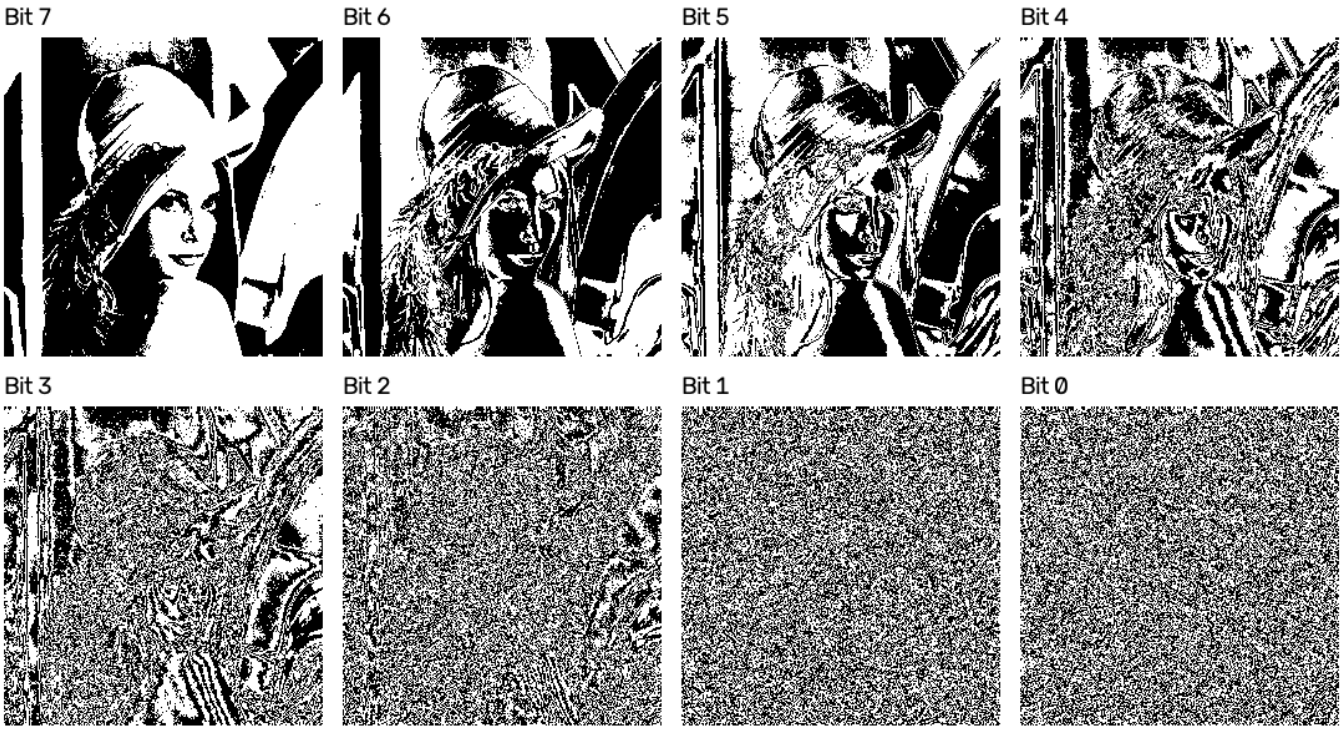
Disgussion

關於這兩張圖的 k 皆為 5，lena 的 PSNR 為 35.7179dB； baboon 為 35.7095dB

3.2
Figure

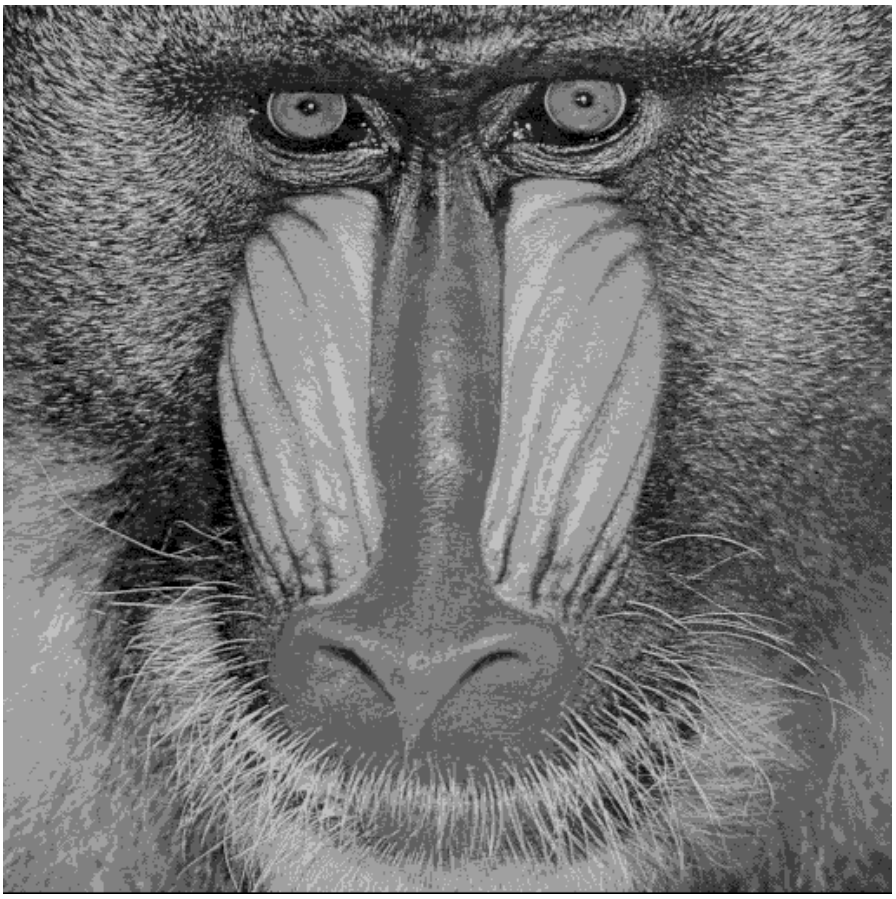


baboon512_all_8_bitplanes.png



lena256_all_8_bitplanes.png

3.2
Figure

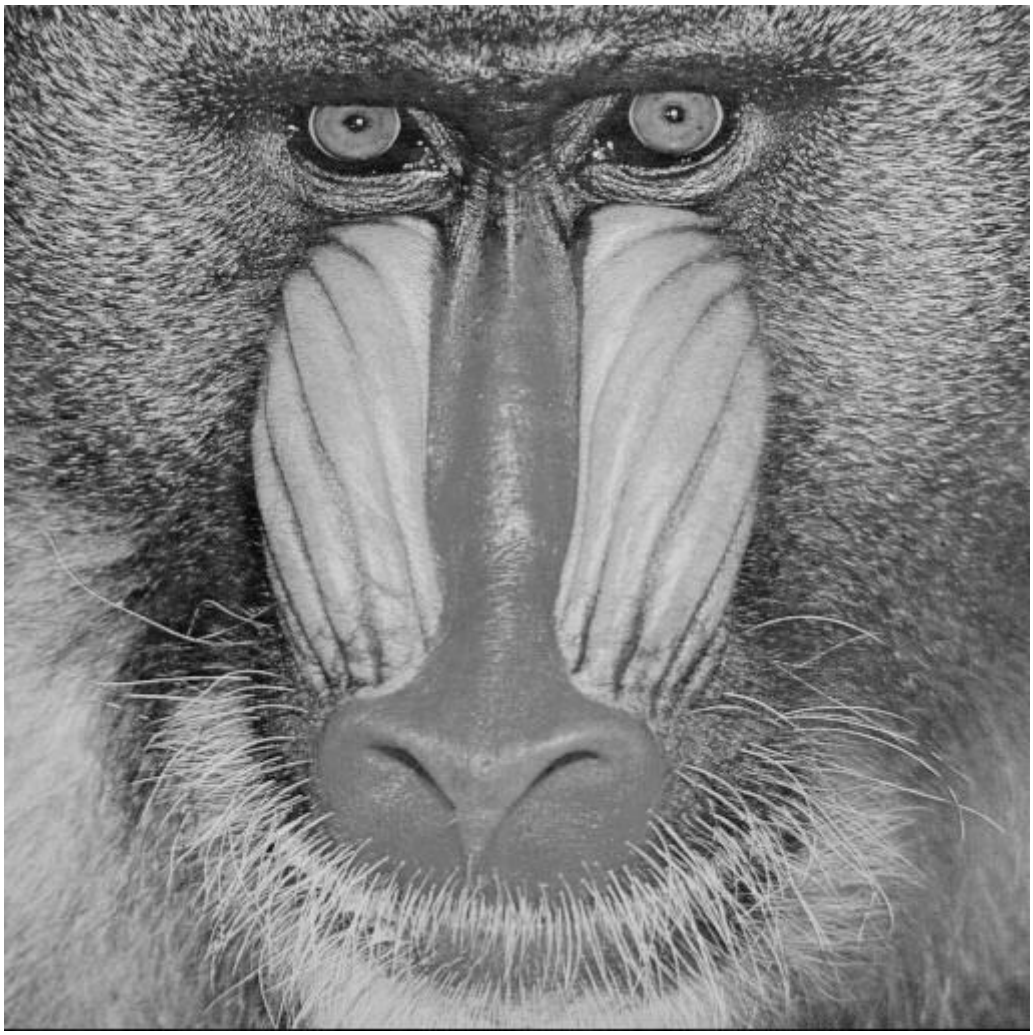


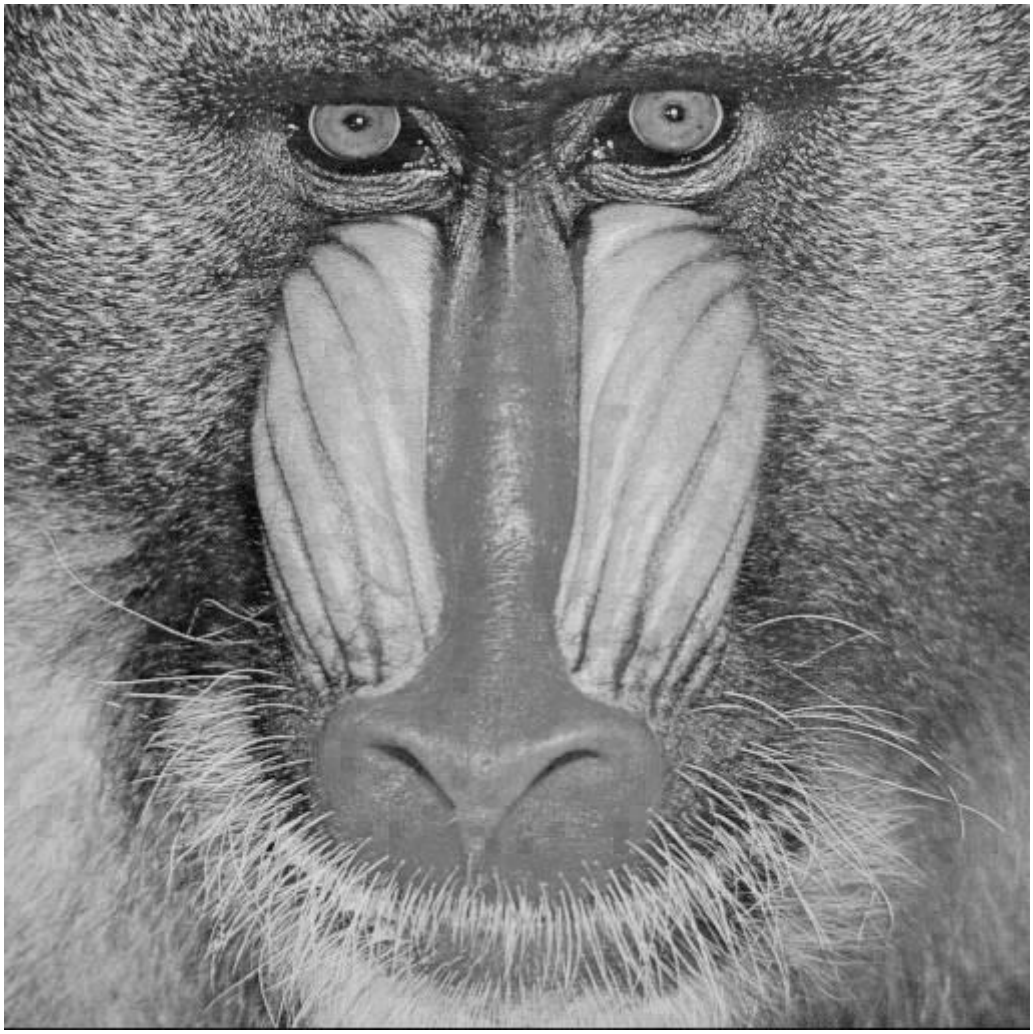
baboon512_structure_reconstructed.png

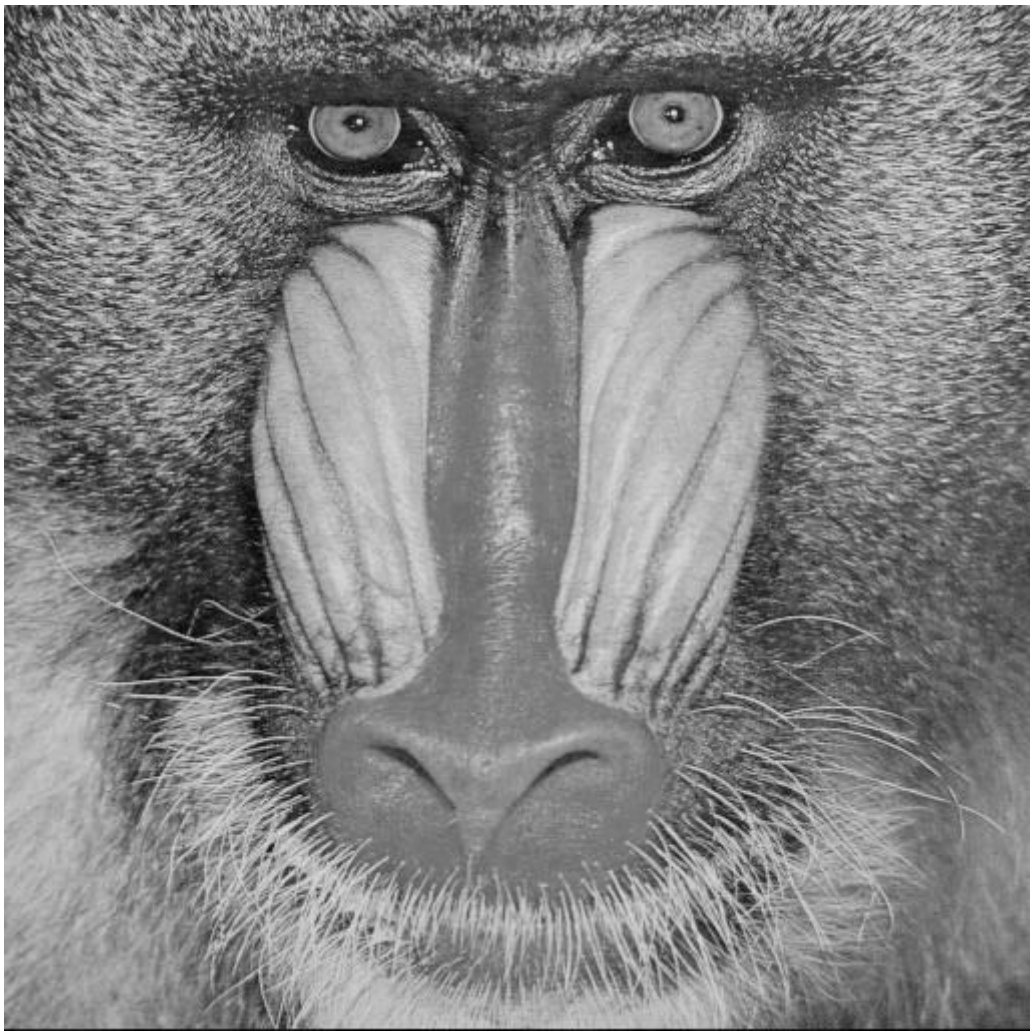


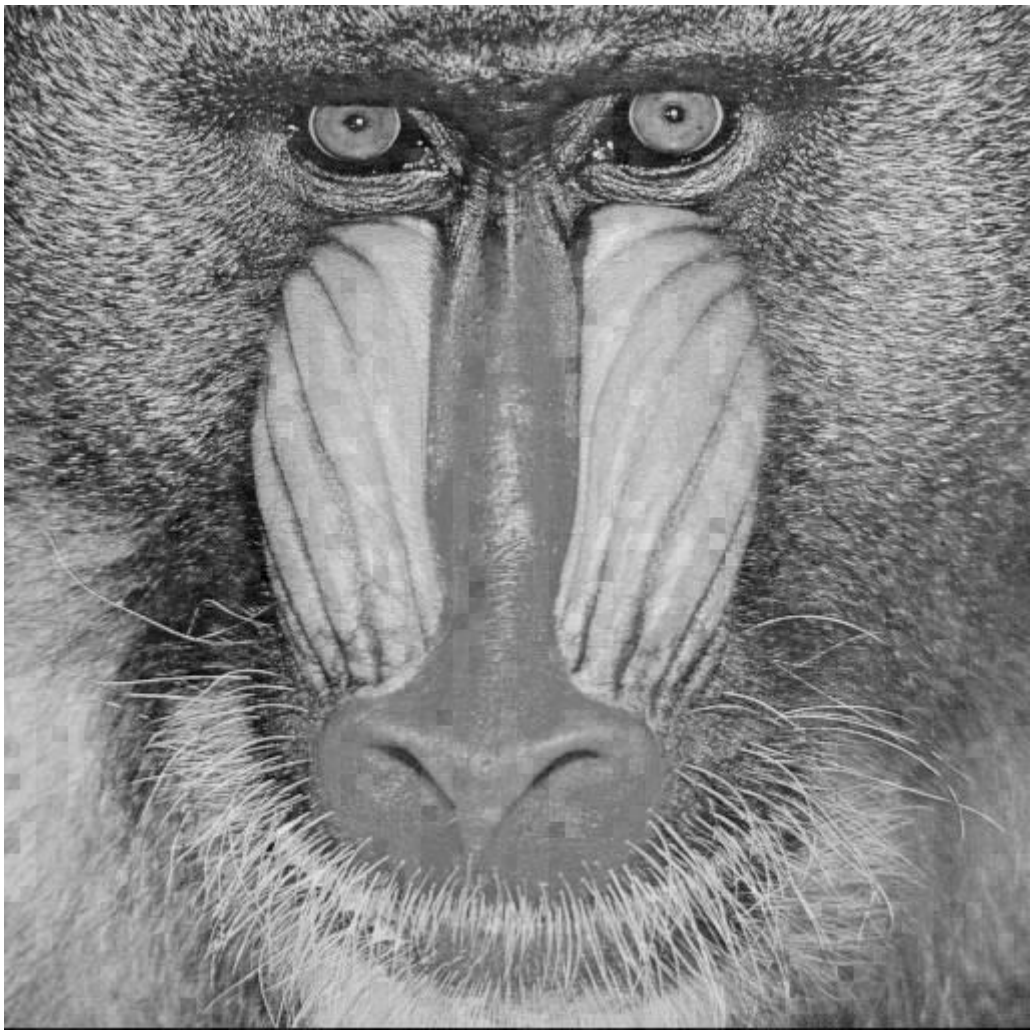
lena256_structure_reconstructed.png

Disgussion
目測的結果顯示 baboon 的毛髮即使 k=4 也還看不太出來有壞掉的問題，反而 lena 會有這方面的問題，當 lena 的 k<5 時，他的皮膚會出現很明顯的斷層，所以最終 lena 維持 k=5, 而 baboon 則可以 k=4

4.1
Figure

baboon512_LSB_encoded.png

4.2
Figure

baboon512_QIM_delta8_encoded.png

4.3
Figure

baboon512_QIM_delta4_encoded.png



baboon512_QIM_delta16_encoded.png

Disgussion

關於 quantization step size，我選了 4 和 16 來實作，從圖片上來看，16 的是可以看出有方塊殘留在圖片上；另外，對於經過 quantization 之後，LSB 全數陣亡，QIM 則很有限的抵禦了，在 step size=4 的狀況中，全部 4096 個區塊中錯了 528 個區塊；在 step size = 16 的狀況中，全部 4096 個區塊中錯了 450 個；如果今天 QIM 的值調整過後落在量化邊界上，在量化的過程中有可能掉下去或跑上去，這樣在解碼的時候就會解出錯誤的數值

另外，老師在上課講解時說的是平均值，但題目寫的是中位數，本題以中位數實作